

Symmetry Breaking in Dependency Quantified Boolean Formulas

Paul Seip

Johannes Kepler University Linz

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(joint work with Clemens Hofstadler, Martina Seidl, Manuel Kauers)

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“Symmetry, as wide or as narrow as you may define its meaning, is one idea by which man through the ages has tried to comprehend and create order, beauty and perfection.”

– Hermann Weyl, 1952

Symmetry $\sim$ Structure

can often be exploited

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A **symmetry** is a *transformation* of an object that leaves a certain property *invariant*.

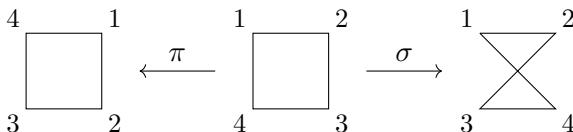


Figure: Transformation of a square by two permutations; only π is a symmetry of the square.

Symmetric formulas (SAT/QBF/DQBF)

Symmetric structure can potentially be exploited for:

- Faster solving (symmetry breaking),
- Shorter refutation proofs (symmetry rule, 'w.l.o.g.').

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Decomposition of the search space into 'symmetry orbits'

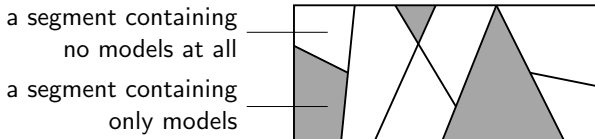


Figure: Search space of a formula Φ

Problem and Goal

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Problem

In SAT and QBF, standard symmetry breaking selects one representative from each symmetry orbit.

In DQBF, this is not automatically sound: a representative constructed orbit-wise may violate the dependency structure.

Goal of this talk

Develop dependency-aware symmetry breaking for DQBF:
shrink the search space while preserving truth.

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$$\Phi = P.\phi = \forall x_1, x_2, x_3 \exists y_1(\{x_2\}), y_2(\{x_3\}).\phi,$$

with *matrix*:

$$\phi = (x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3).$$

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DQBF Models

$$\Phi = \forall x_1, x_2, x_3 \exists y_1(\{x_2\}), y_2(\{x_3\}). [(x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3)]$$

Skolem model: $y_1 = f_1(x_2) = x_2$, $y_2 = f_2(x_3) = x_3$.

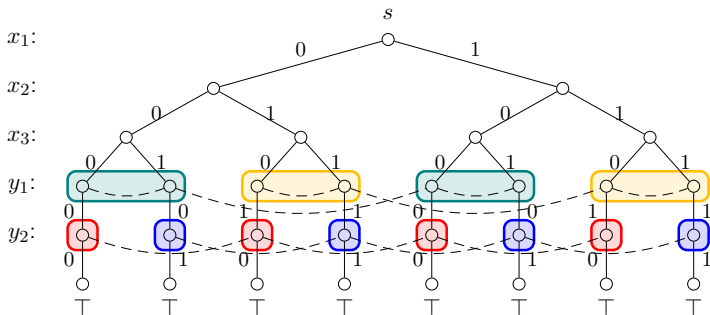


Figure: Model of Φ , with coloured dependency clusters.

Syntactic Symmetry - Example

$$\Phi = \forall x_1, x_2, x_3 \exists y_1(\{x_2\}), y_2(\{x_3\}). [(x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3)]$$

$$\Phi = P.\phi = (x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3)$$

$$\sigma(\Phi) = P.\sigma(\phi) = P.(x_1 \leftrightarrow y_1) \vee (\neg x_2 \leftrightarrow \neg y_2) \vee (x_1 \leftrightarrow x_3)$$

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Induced Semantic Symmetry

$$\Phi = \forall x_1, x_2, x_3 \exists y_1(\{x_2\}), y_2(\{x_3\}). [(x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3)]$$

$$y_1 = f_1^\sigma(x_2) = \neg x_2, \quad y_2 = f_2^\sigma(x_3) = \neg x_3.$$

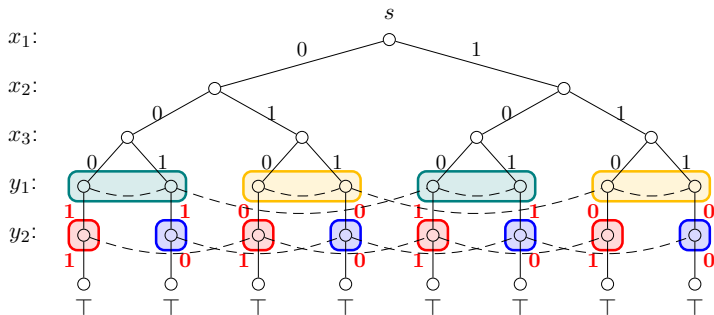


Figure: Symmetric model f^σ of Φ

Symmetry Breakers

Fix a DQBF $\Phi = P.\phi$.



Idea: shrinking the search space by excluding 'symmetric interpretations'.

Conjunctive Symmetry Breaker

A *conjunctive symmetry breaker* for a symmetry group G_Φ is a propositional formula ψ with the property that

$$P.\phi \text{ is true} \iff P.(\phi \wedge \psi) \text{ is true,}$$

ψ shrinks the search space while keeping at least one representative from each symmetry orbit.

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Lex-leader Symmetry Breaking

$$\text{leq}(X, Y) = X \leq Y := \bigwedge_{i \in [0, n-1]} \left(\left(\bigwedge_{j \in [0, i-1]} (x_j = y_j) \right) \rightarrow (x_i \leq y_i) \right).$$

ψ_{lex} selects interpretations from orbits based on a lexicographic ordering of the search space.

$$\psi_{\text{lex}} = \bigwedge_{\pi \in G_{\text{sym}}} \text{leq}(X, X^{\pi}).$$

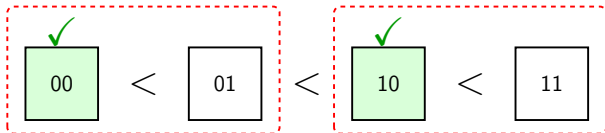


Figure: Example of lex-leader symmetry-breaking in SAT, ψ_{lex} eliminates 01 and 11.

Lex-leader Symmetry Breaking for DQBF

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DQBF obstruction

Naive lex-leaders may eliminate all valid Skolem models.

$$\Phi = \forall x_1, x_2, x_3 \exists y_1(\{x_2\}), y_2(\{x_3\}). [(x_1 \leftrightarrow y_1) \vee (x_2 \leftrightarrow y_2) \vee (x_1 \leftrightarrow x_3)]$$

Symmetry σ : $x_2 \Leftrightarrow \neg x_2$, $y_2 \Leftrightarrow \neg y_2$

$$\begin{aligned}\psi_{\text{lex}} &= (x_2 \leftrightarrow \sigma(x_2)) \rightarrow (y_1 \rightarrow \sigma(y_1)) \wedge \\ &\quad (x_3 \leftrightarrow \sigma(x_3) \wedge y_1 \leftrightarrow \sigma(y_1)) \rightarrow (y_2 \rightarrow \sigma(y_2)) \\ &\equiv \neg y_2.\end{aligned}$$

But $P.(\phi \wedge \neg y_2)$ is **false**.

Lex-leader Symmetry Breaking for DQBF

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But $P.(\phi \wedge \neg y_2)$ is **false**.

Combining Symmetries

Problem: combining symmetries may not interact well with the dependency structure.

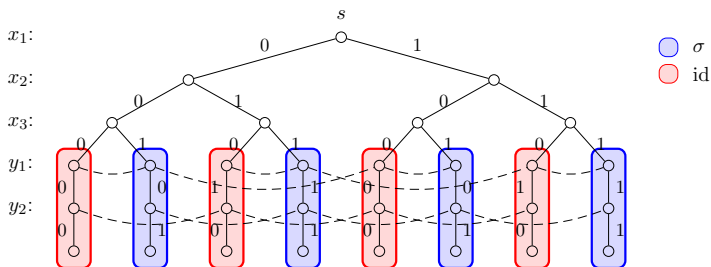


Figure: A mixed symmetry acting on f that is not well-defined

Solution Sketch

Two ingredients:

- i A **gluing condition** w.r.t. the dependency structure in the mixing theorem;
- ii A **nonlinear dependency condition**. For every incomparable pair D_i, D_j , whenever g acts nontrivially on y_i , it fixes y_j and fixes every universal variable in $D_j \setminus D_i$.

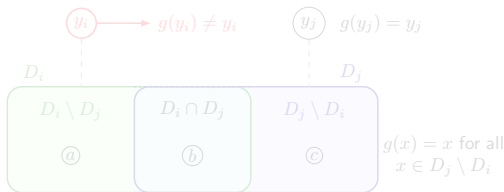


Figure: An illustration of the nonlinear dependency condition

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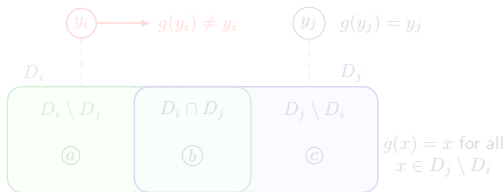


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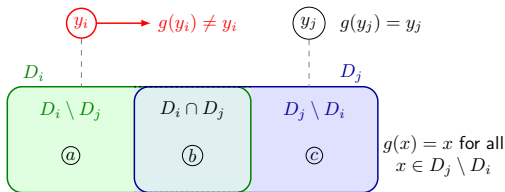


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- The symmetry σ in our example does not satisfy the nonlinear dependency condition.
- **Intuition:** incomparable dependency sets can force contradictory gluing requirements.
- Remark: the condition appears restrictive, but for linear dependency structures (QBF), the condition vanishes and we recover the lex-leader symmetry breaking construction for QBF.

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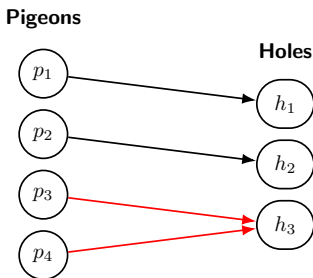
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Pigeonhole Principle

$\text{PHP}_n := \exists \text{ injective map } \{1, \dots, n+1\} \rightarrow \{1, \dots, n\}.$



Example: PHP_3

Classic symmetry-rich benchmark in propositional logic.

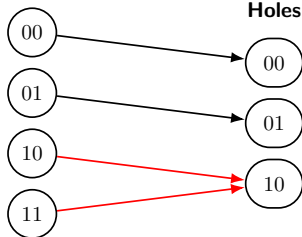
Dependency Pigeonhole Principle



Idea: try to encode the false claim

$$\text{DPHP}_n := \exists \text{ injective map } \{0,1\}^n \rightarrow \{0,1\}^n \setminus \{1^n\}.$$

Pigeons



Example: DPHP_2

Can naturally be encoded as a DQBF – needs incomparable dependency sets.

DPHP - Symmetries

- ‘Pigeon symmetries’: simultaneous literal permutations in $U, V - 2^n n!$,
- ‘Hole symmetries’: simultaneous literal permutations in $P, Q, H - 2^n n!$,
- ‘Copy-swap symmetry’: simultaneously swap $U \Leftrightarrow V, P \Leftrightarrow Q$.

$$|G_{\text{sym}}^{\Phi}| = 2(2^n n!)^2.$$

DPHP - Symmetries

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Permissible Symmetries for Symmetry Breaking

Permissible for SB

- | | |
|------------------------|---|
| (1) Pigeon symmetries | ✓ |
| (2) Hole symmetries | ✗ |
| (3) Copy-swap symmetry | ✗ |
-

We use the subgroup $|G_{(1)}| = 2^n n!$ for symmetry breaking.

Permissible Symmetries for Symmetry Breaking

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We use the subgroup $|G_{(1)}| = 2^n n!$ for symmetry breaking.

Preliminary Experimental Results - DPHP_n

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Runtime (s) - Solver Pedant		
n	Without symmetries	With symmetries
3	0.17	0.01
4	T/O	0.02
5	T/O	0.02
10	T/O	0.03
20	T/O	0.07

T/O: timeout after 30 seconds.

Conclusion

- DQBF symmetry breaking is more delicate than SAT/QBF symmetry breaking because dependencies may be incomparable.
- Naive lex-leader lifting is not sound.
- Dependency-aware restrictions recover sound symmetry breaking.
- Symmetry-rich benchmarks such as DPHP still contain useful permissible symmetry which can significantly speed up solving time.

Thank you! ... Questions?

